MCO Wave Parameter Library

ADDENDUM — Taxi Modes and Park Mode · 2026-06-03

1. Taxi Modes — Ground Locomotion

The MCO-1D can propel itself along the ground at walking to jogging speed using very low amplitude flapping that stays within the undercarriage clearance envelope. Turning uses differential amplitude between port and starboard wing pairs, validated by peer-reviewed ornithopter research. Reversing uses a negative pitch offset to tilt the thrust vector rearward.

1.1 Literature basis

Lilong & Yu (2024) Biomimetics 9(3):148 — bilateral amplitude asymmetry ΔΦ* in the range 0–0.4 produces a coupled roll+yaw moment, with roll and yaw in a linear relationship. Efficient for directional control without a tail rudder. ΔΦ* = (Φ_high – Φ_low) / Φ_mean. On the ground the undercarriage resists roll, leaving the yaw component dominant — exactly what is needed for ground turning. Patent US7121505 validates asymmetric ornithopter flapping for turning. Science.gov wing stroke amplitude review confirms yaw torque is independently controlled by differential amplitude.

1.2 Geometry constraint

At ground level the wing tips must clear the ground surface. With H_GEAR = 200mm and a 150mm safety margin:

Parameter	Value	Calculation
Max safe amplitude at rest	12°	arccos(1 – 50/712) = 11.3° → 12° with margin
Ground clearance at 12°	156mm	712×(1–cos12°) = 15.5mm inward; tip 156mm above ground
Taxi amplitude	12°	Used for all taxi modes — never exceeds this
Reverse amplitude	10°	Reduced for extra margin during rearward movement

1.3 Taxi mode parameters

Parameter	Value	Notes
TAXI_FREQ_SLOW_HZ	0.3 Hz	Walking pace ~0.5 m/s ground speed
TAXI_FREQ_FAST_HZ	0.6 Hz	Jogging pace ~1.2 m/s ground speed
TAXI_AMP_DEG	12°	Max safe at ground level — 150mm clearance margin
TAXI_PITCH_DEG	+2°	Slight positive AoA — ground effect augments thrust
TAXI_TURN_DELTA_PHI	0.30	Asymmetry ratio ΔΦ*; range 0–0.4 per literature
TAXI_REVERSE_FREQ_HZ	0.4 Hz	Rearward movement frequency
TAXI_REVERSE_AMP_DEG	10°	Reduced amplitude for reverse safety
TAXI_REVERSE_PITCH_DEG	–12°	Negative pitch tilts thrust vector rearward
Phase offset (all taxi)	90°	Standard metachronal cascade unchanged
Sweep angle (all taxi)	60°	All wings fully deployed for maximum ground clearance

1.4 Turning mechanism

Manoeuvre	Port amplitude	Starboard amplitude	Notes
Straight ahead	12°	12°	Symmetric — no yaw torque
Gentle right	13.8°	10.2°	ΔΦ*=0.30 at full yaw stick
Gentle left	10.2°	13.8°	ΔΦ*=0.30 at full yaw stick
Pivot right (spot turn)	12°	0°	Max asymmetry — aircraft spins in place
Reverse straight	10°	10°	Negative pitch; same steering rules apply
Reverse right	10.8°	9.2°	Steering while reversing

1.5 FCC integration

Taxi mode activates when: armed AND throttle > 0 AND ground proximity state is 'ground' or 'ground_roll'. Reverse activates via a dedicated switch (CH7 suggested). Taxi mode is suspended when the FCC detects liftoff (AGL > 0.15m or airspeed > 3 m/s).

2. Park Mode — Post-Flight Check and Secure

Park mode runs after landing, before the battery is disconnected. It mirrors the pre-flight protocol but focuses on what the flight may have changed: battery state, servo health after vibration, motor condition, and pilot-prompted physical inspection.

2.1 Aviation standard basis

AOPA post-flight procedures: after landing the pilot should confirm whether the aircraft has any mechanical problems needing attention before the next flight, and advise maintenance. **CMU UAS post-flight checklist**: hull, propellers, computers, power, data download, batteries within safe storage voltage. **Standard aviation post-flight**: control surfaces inspected for freedom, damage, and wear. Engine/power system inspected for leaks, overheating, unusual sounds. Log any discrepancies.

2.2 Five-step park sequence

Step	Duration	What is checked	Blocks next flight if?
1 Electronics	~15s	Battery landing voltage vs thresholds. Flight anomaly log summary. CT measurement recorded.	Battery <19.8V
2 Servo health	~30s	Each sweep servo exercised 60°→120°→60°. Timing checked for sluggishness (rope binding = wear flag).	Servo fails to respond
3 Motor health	~10s	All 8 motors spun at 0.2 Hz for 2s with wings stowed. Unusual sounds or CAN faults logged.	Motor fails to respond
4 Final sensors	~5s	Airspeed at rest (should be <3 m/s). Attitude (should be near level). Logged.	Airspeed >3 m/s (blocked pitot)
5 Physical inspection	~2 min	Pilot confirms: membranes, D-box skin, rope drives, fasteners, wiring, connectors, battery connector.	Any 'fix' response from pilot.
Stow	~20s	W4→W3→W2→W1 progressive stow to 210° at 20°/s.	—

2.3 Output

Park mode writes a park log file: flight_NNN_DATETIME_park.txt. Contains battery landing voltage, anomaly count, issue list, and clear-for-next-flight status. This file is included when pasting data to Claude for between-session analysis.

3. Wave Library Parameter Block

Add the following to wave_library.py alongside existing parameters:

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# ■■ TAXI MODE ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
TAXI_FREQ_SLOW_HZ = 0.3 # Hz slow taxi ~0.5 m/s
TAXI_FREQ_FAST_HZ = 0.6 # Hz fast taxi ~1.2 m/s
TAXI_AMP_DEG = 12.0 # deg max at ground level
TAXI_PITCH_DEG = +2.0 # deg ground effect AoA
TAXI_TURN_DELTA_PHI = 0.30 # ΔΦ* asymmetry ratio
TAXI_REVERSE_FREQ_HZ = 0.4 # Hz
TAXI_REVERSE_AMP_DEG = 10.0 # deg
TAXI_REVERSE_PITCH_DEG = -12.0 # deg rearward thrust
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